Modern Approaches to Prediction Modeling: Integrating AI and Machine Learning

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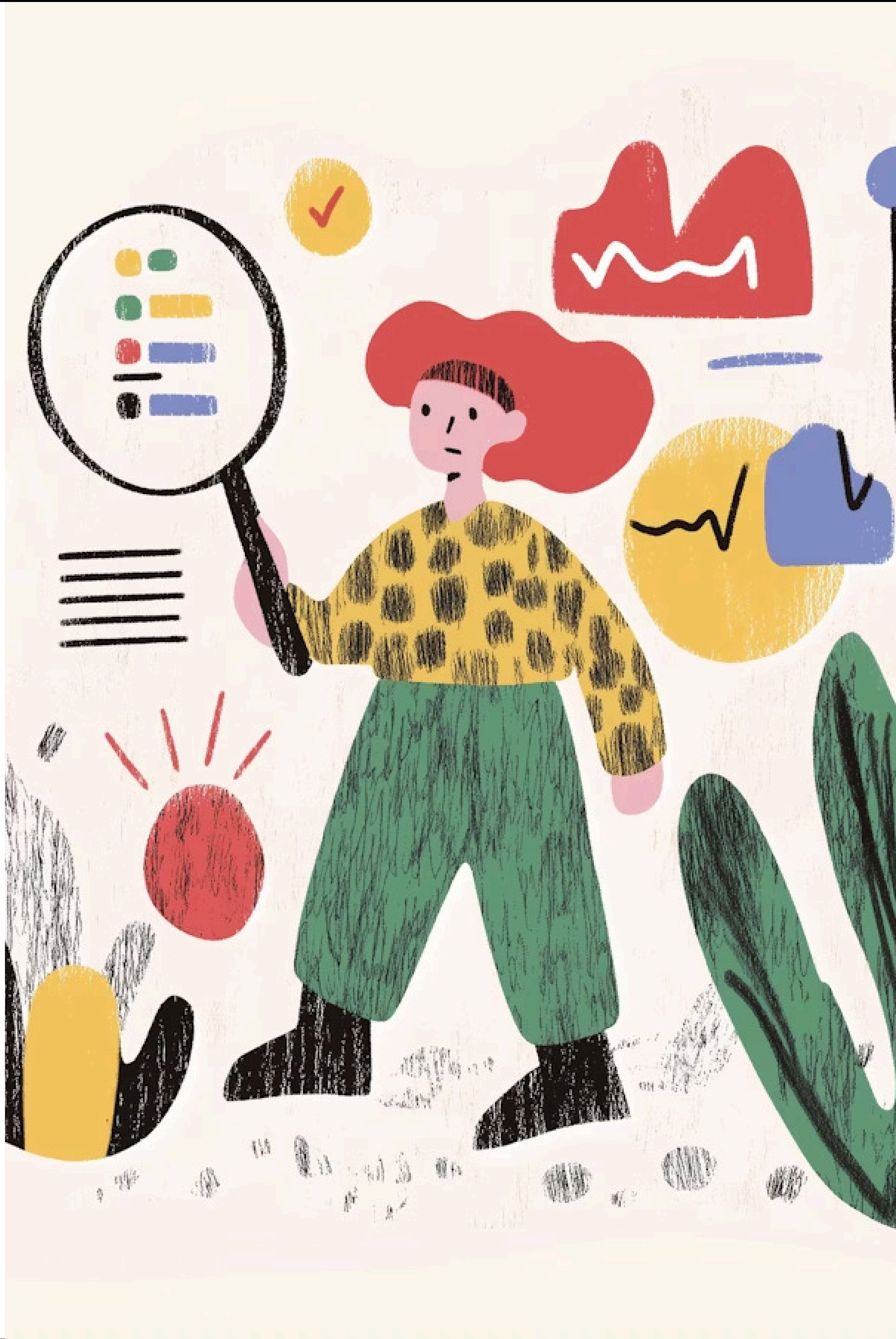


The Evolution: From Association to Prediction

Traditional Association

Research Question:
"Is gene X associated with disease Y?"

Primary Goal:
Causal inference and etiologic understanding

Key Metrics:
P-values, Odds Ratios, Confidence Intervals

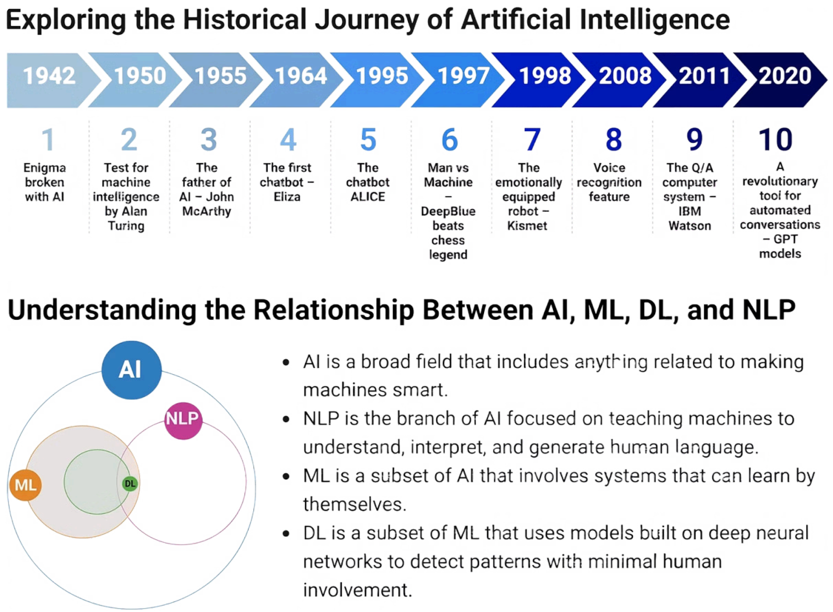
Prediction Modeling

Research Question:
"What is this specific patient's risk of disease Y?"

Primary Goal:
Accurate risk stratification and clinical decision support

Key Metrics:
Accuracy, AUC, Calibration, PPV/NPV, NRI

Today's data landscape is extraordinarily complex: EHR systems, multi-omics platforms, natural language processing, and advanced imaging all generate massive datasets. Our thesis: **The most powerful models integrate AI innovation with biostatistical rigor**, rather than relying on either approach alone.



The Evolving Synergy: Biostatistics Meets AI

1. **Biostatistics Foundation**

   Provided a rigorous foundation for medical research, focusing on causal inference, hypothesis testing, and understanding disease mechanisms.

2. **AI's Rise & Capabilities**

   Emergence grew with data volume and computational power, excelling in pattern recognition, predictive modeling, and complex data integration.

3. **Synergistic Impact**

   Integrating biostatistics' rigor with AI's predictive power yields impactful advancements, leading to more comprehensive, reliable, and clinically actionable health insights.

Alowais et al. Revolutionizing healthcare: the role of artificial intelligence in clinical practice. BMC Medical Education, 2023

Challenge #1: Defining Reliable Outcomes

Before building prediction models, we must establish trustworthy outcome labels.

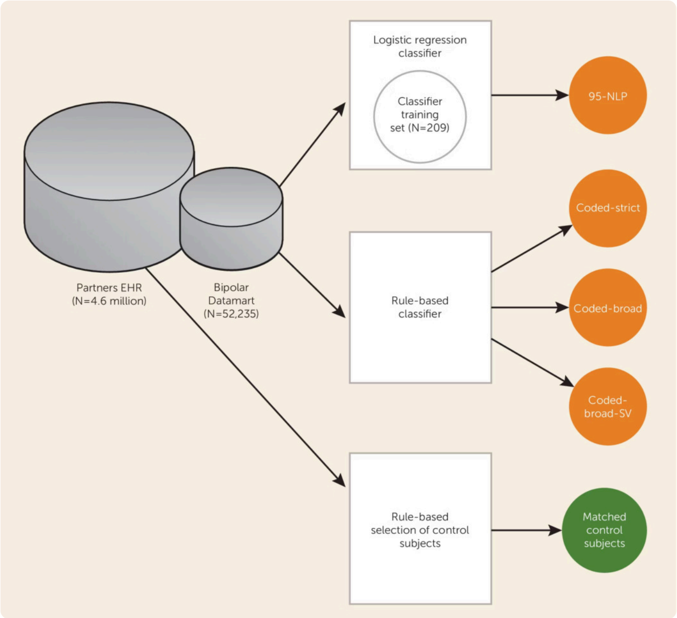
Phenotypes derived from insurance billing codes often lack the precision needed for robust outcome definition, introducing measurement error and bias into prediction models.

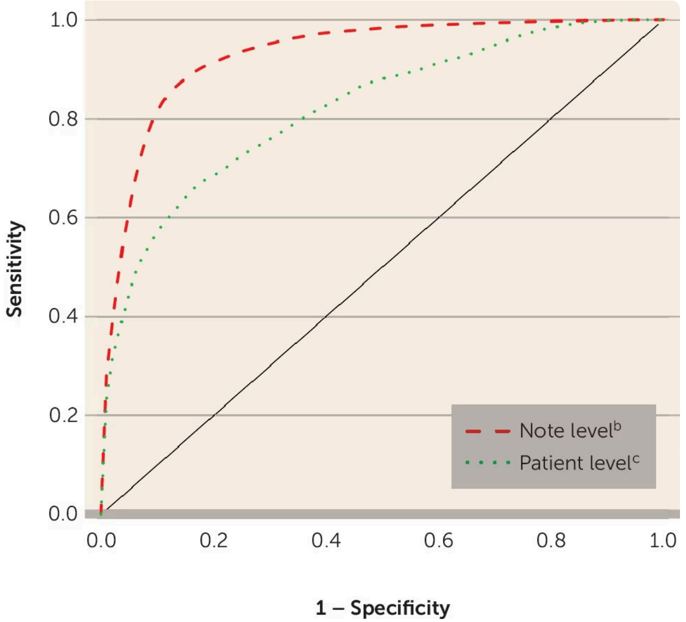
While a traditional gold standard, manual chart review is a labor-intensive process, creating a significant bottleneck that hinders research scalability and efficiency.

Machine learning and advanced statistical methods offer promising avenues to systematically define and validate reliable outcome labels, overcoming current limitations.

How can these innovative methods help us accurately define what we're trying to predict?

Case Study 1: Phenotyping Bipolar Disorder with Natural Language Processing





Research Goal

Identify Bipolar Disorder cases accurately within electronic health records

Methodological Approach

Applied Adaptive LASSO (regularized regression) combining coded diagnostic data with NLP-extracted features from clinical notes

Validation Strategy

Compared algorithm predictions against gold standard SCID (Structured Clinical Interview) assessments

Clinical Impact

Achieved PPV of 0.85, enabling creation of a reliable research cohort for subsequent genetic studies (4500 cases, 5000 controls)

Castro, Minnier et al. Validation of electronic health record phenotyping of bipolar disorder cases and controls. American Journal of Psychiatry, 2015

Case Study 2: Rheumatoid Arthritis with Limited Labels

The Innovation

Challenge: Automated feature selection with minimal labeled data

Solution: Unsupervised clustering on surrogate markers (ICD code counts) to generate "silver standard" labels, followed by Adaptive LASSO selection

Result: Achieved equivalent accuracy using only 50% of the labeled training data

Gronsbell, Minnier et al. Automated feature selection of predictors in electronic medical records data. Biometrics, 2018

Key Takeaway: Machine learning and statistical methods are essential not just for prediction, but for defining the very outcomes we seek to predict.

Challenge #2: Integrating Complex, Multi-Modal Data

Once reliable outcomes are established, the next frontier involves modeling complex, high-dimensional predictors from diverse data sources. How do we effectively integrate genomic data, clinical variables, imaging features, and unstructured text?

Case Study 3: Non-Linear Genomic Risk Modeling

1

The Problem

Disease risk from thousands of SNPs exhibits non-linear relationships and gene-set structures that simple linear models cannot capture

2

The Solution

Adaptive Naive Bayes Kernel Machine (ANBKM)

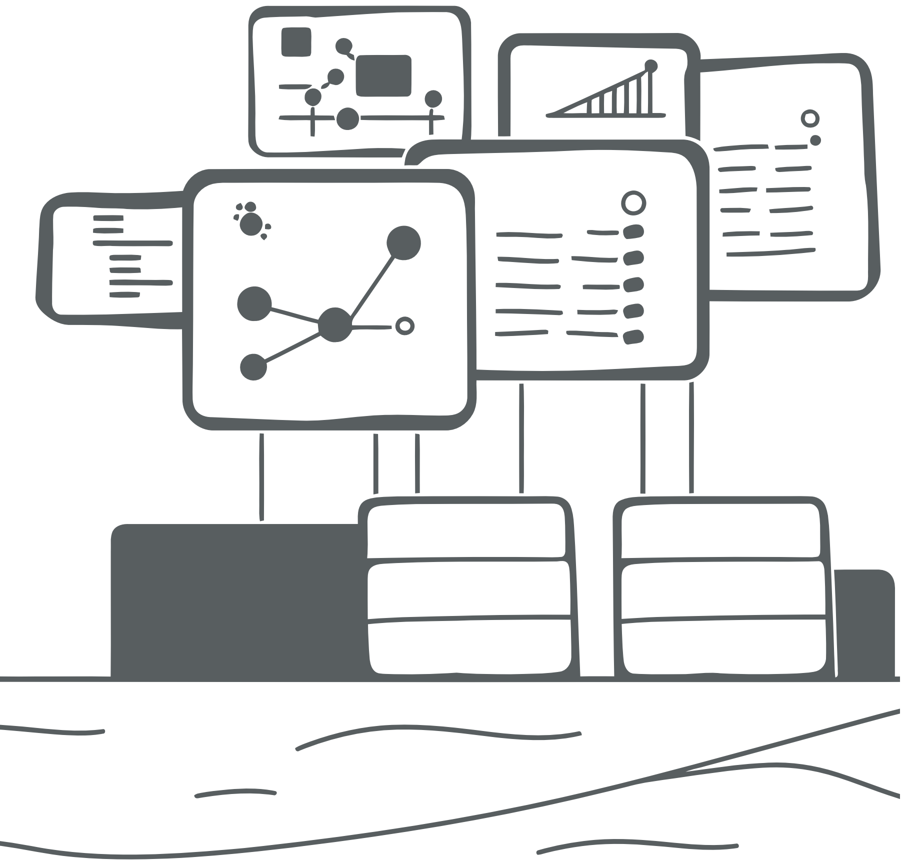
- Kernel Machine: Models complex, non-linear risk within gene-sets
- Naive Bayes + LASSO: Aggregates risk scores and selects informative gene-sets

3

The Advantage

Captures biological pathway structures while maintaining statistical interpretability and feature selection

Minnier et al. Risk classification with an adaptive naive Bayes kernel machine model. Journal of the American Statistical Association, 2015



Case Study 4: Ensemble Learning for Health Equity

Research Context

Goal: Infer Latino nativity status for health equity research using diverse EHR data sources

Data Sources:

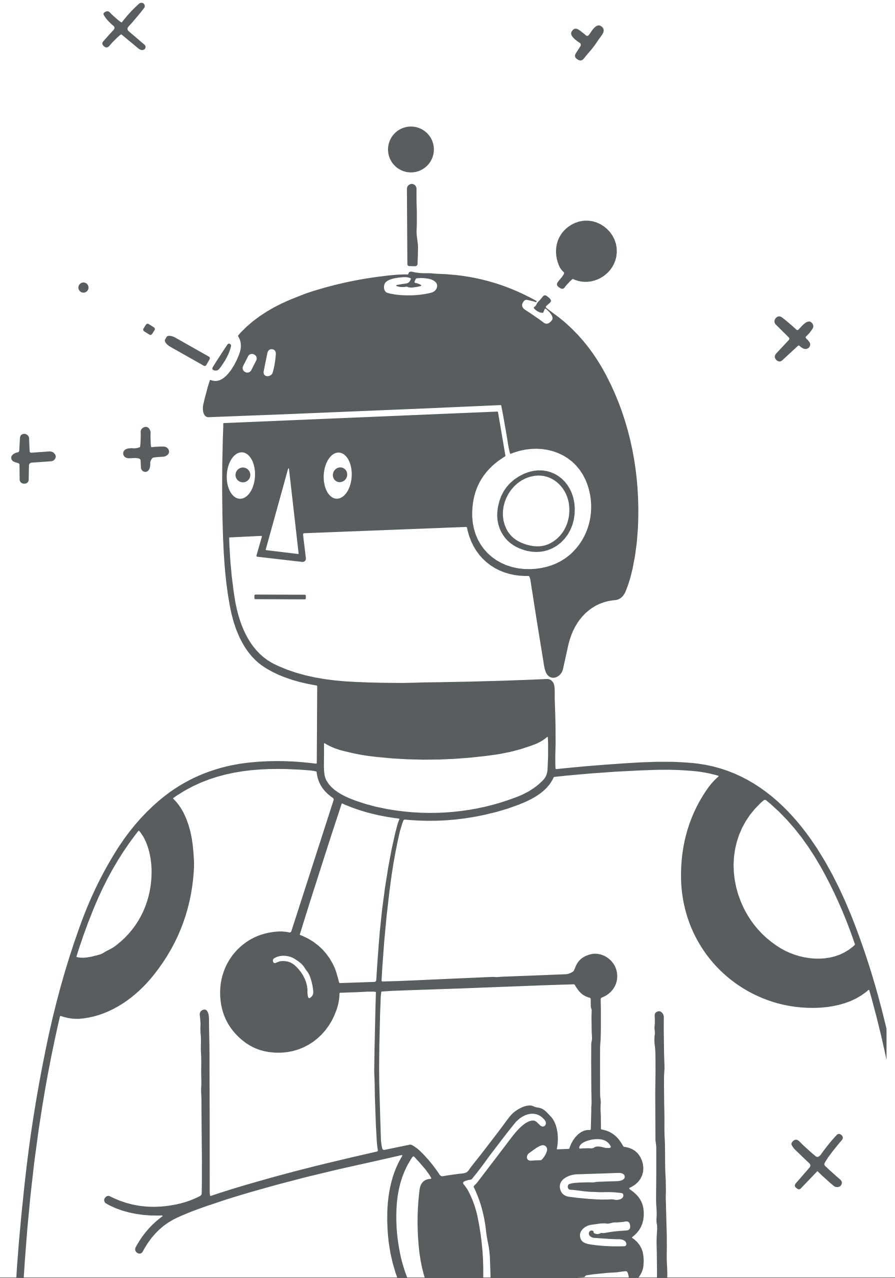
- Electronic health records
- Census tract information
- Surname-based indicators

Super Learner Approach

Combined multiple algorithms (Logistic Regression, Random Forest, Gradient Boosting) using optimal weighted averaging

Performance: Ensemble model achieved AUC > 0.90, substantially outperforming any individual algorithm

Marino, Minnier et al. Health Services Research, 2023



The Frontier: Integrating "Black Box" AI into Prediction Models

The New Reality

FDA-authorized AI algorithms now provide powerful, stand-alone predictions for clinical use

The Biostatistician's Role

Our challenge: How do we rigorously integrate these AI tools with other predictive data sources?

The Opportunity

Create hybrid models that leverage AI strengths while maintaining statistical interpretability

Project Spotlight: VA Breast Cancer Risk Integration

We are developing the most comprehensive breast cancer risk model by integrating all available predictive data sources.

Clinical Data

Age, family history, reproductive factors, breast density, etc.

Genomics

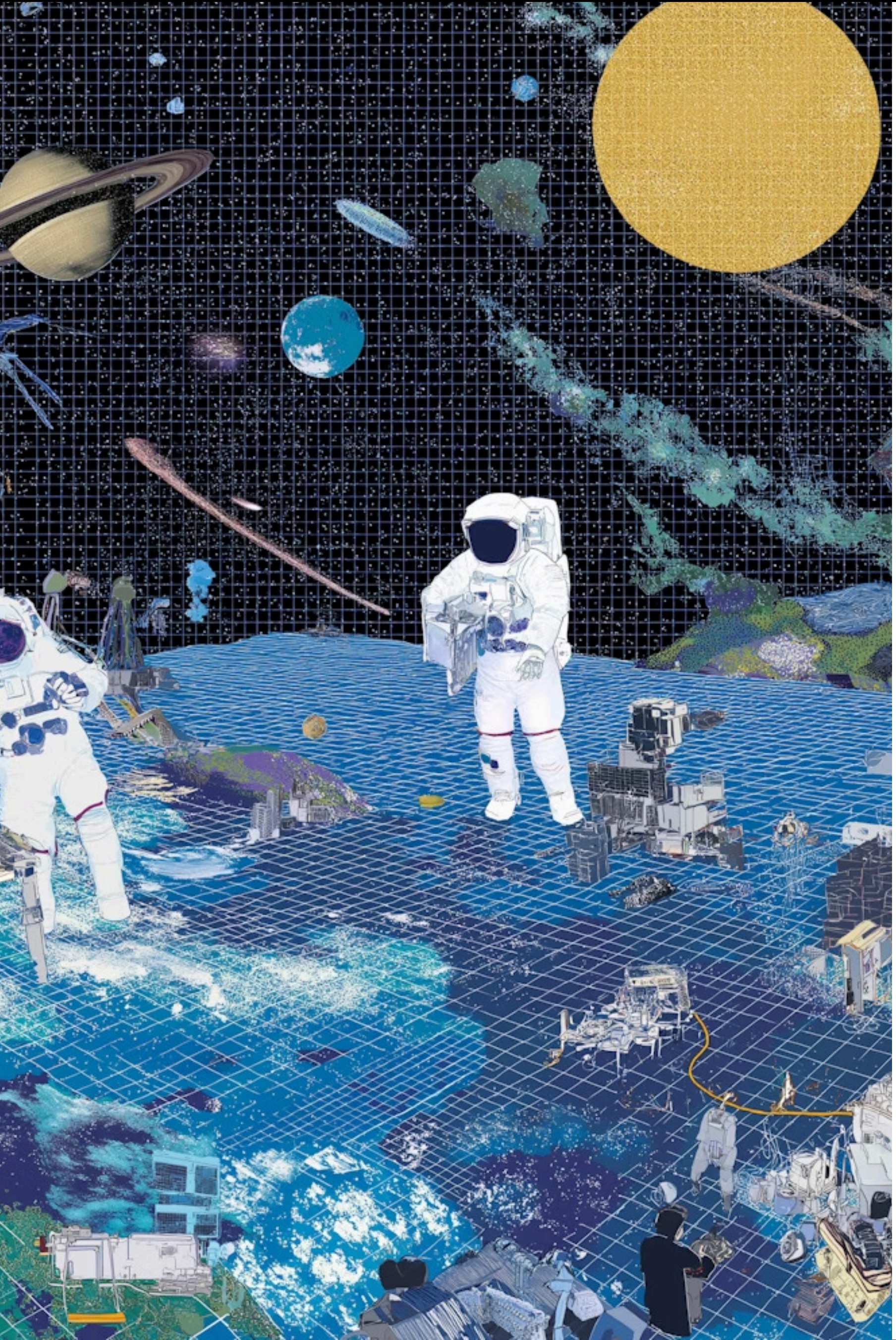
Polygenic Risk Score (PRS) from genome-wide association studies

AI Imaging

"Clairity" algorithm score: deep learning model analyzing mammogram patterns

Statistical Innovation: New_Risk_Model = f(Clinical_Data, PRS, AI_Score)
Don't just compare AI to genomics—integrate them for maximum predictive power.

Clinical Impact: Identify high-risk women (elevated PRS + high AI score) for enhanced screening protocols and prevention interventions.



The Next Frontier: Large Language Models for Feature Extraction

Clinical notes contain rich information absent from structured fields: nuanced social determinants of health, detailed family histories, symptom severity trajectories, and treatment response patterns. LLMs represent the next generation of NLP, capable of extracting this complex information at scale.

80%

Unstructured Data

Percentage of EHR data in free-text format

From Traditional NLP to LLM-Powered Extraction

Traditional NLP Approach

Our Previous Work:

- Bipolar Disorder: Keyword detection ("racing thoughts", "grandiosity")
- Rheumatoid Arthritis: Pre-defined term extraction
- Manual feature engineering required

Limitations: Labor-intensive, limited to predefined concepts

LLM-Powered Future

Emerging Capabilities:

- Automated reading of entire clinical narratives
- Complex concept extraction without manual rules
- Contextual understanding of medical language

Advantage: Scales to millions of notes, captures nuanced clinical patterns

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Example Research Question: Can LLMs automate extraction of Bipolar subphenotypes (psychosis features, suicide attempts, age of onset) that we previously validated through manual review? If successful, this creates powerful new predictors for risk stratification models.

Key Conclusions: The Prediction Modeling Pipeline

1

1. Rigorous Phenotyping

Use ML and statistical methods to define reliable outcome labels when gold standards are limited

2

2. Flexible Modeling

Integrate diverse data types (clinical, genomic, imaging) using appropriate statistical and ML frameworks

3

3. AI Integration

Thoughtfully combine black-box AI predictions with interpretable statistical models for maximum clinical utility

4

4. Rigorous Validation

Assess calibration, clinical utility, and real-world performance—not just AUC

Critical Validation and Implementation Challenges

Beyond AUC: Comprehensive Validation

- **Calibration:** Do predicted 30% risks match observed 30% event rates?
- **Clinical Utility:** Does the model change clinical management decisions?
- **PPV/NPV:** What is real-world predictive performance in target populations?

Power Analysis and Grant Strategy

Calculate sample size requirements for *incremental value* metrics:

- ΔAUC improvement from adding AI component
- Net Reclassification Improvement (NRI)
- Feature variability estimation: Large sample sizes needed for stable estimation in models with many predictors
- Often requires simulation studies to demonstrate feasibility

Funding and Implementation

- **Funding Sources:** Pursuing NIH and VA grants for validation and implementation
- **Key Requirements:**
  - Access to images and/or non-coded text fields
  - Ability to implement AI models in clinical workflows
  - Clinical trial studies examining utility of screening or early diagnosis

Thank You

Questions?

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